

Hall-sensor-based hexapole magnetic flux model: $D\mathbf{v}_j = \frac{k_m}{\hat{\ell}^2} \hat{\mathbf{Q}}_j$ for $j = 1, 2, \dots, N_I$

$$\mathbf{b}_{ij} = \mathbf{S}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) D\mathbf{v}_j$$

$$\mathbf{S}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) = \begin{bmatrix} \underbrace{\frac{\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{ck}}{\|\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{ck}\|^3}}_{s_{i1}} & \underbrace{\frac{\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{ck}}{\|\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{ck}\|^3}}_{s_{i2}} & \dots & \underbrace{\frac{\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{ck}}{\|\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{ck}\|^3}}_{s_{i6}} \end{bmatrix}$$

Model parameter estimation:

$$\boldsymbol{\varepsilon}_{ij}(\mathbf{d}) = \mathbf{b}_{ij} - \mathbf{S}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \underbrace{D\mathbf{v}_j}_{\mathbf{d}_j^v},$$

$$\mathbf{e}^T = \begin{bmatrix} \underbrace{e_{1x} \ e_{1y} \ e_{1z}}_{\mathbf{e}_1^T} & \underbrace{e_{2x} \ e_{2y} \ e_{2z}}_{\mathbf{e}_2^T} & \underbrace{e_{3x} \ e_{3y} \ e_{3z}}_{\mathbf{e}_3^T} & \underbrace{e_{4x} \ e_{4y} \ e_{4z}}_{\mathbf{e}_4^T} & \underbrace{e_{5x} \ e_{5y} \ e_{5z}}_{\mathbf{e}_5^T} & \underbrace{e_{6x} \ e_{6y}}_{\mathbf{e}_6^T} \end{bmatrix}$$

$$\mathbf{e}_6^T = [e_{6x} \ e_{6y} \ (e_{1x} - e_{2x} + e_{3y} - e_{4y} + e_{5z})]$$

where $\mathbf{S}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) = [s_{i1} \ s_{i2} \ \dots \ s_{i6}]$, $\mathbf{v}_j = [v_{j1} \ v_{j2} \ \dots \ v_{j6}]^T$, and $\mathbf{D} \approx \text{diag}(d_1 \ d_2 \ \dots \ d_6)$.

$$\mathbf{d}_j^v = \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \mathbf{S}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \right)^{-1} \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \mathbf{b}_{ij} \right)$$

$$\boldsymbol{\varepsilon}_{ij}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) = \mathbf{b}_{ij} - \mathbf{S}_i(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \mathbf{S}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \right)^{-1} \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \mathbf{b}_{ij} \right),$$

$$J(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) = \sum_{j=1}^{N_I} \left\{ \sum_{i=1}^{N_p} \mathbf{b}_{ij}^T \mathbf{b}_{ij} - \sum_{i=1}^{N_p} \mathbf{b}_{ij}^T \mathbf{S}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \mathbf{S}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \right)^{-1} \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \mathbf{b}_{ij} \right) \right\}$$

Minimize $J(\hat{\ell}, \hat{\mathbf{e}}) \Rightarrow \hat{\ell}, \hat{\mathbf{e}}$ (Initial guess: $\hat{\ell} = \ell_{\text{design}}$ and $\hat{\mathbf{e}} = \mathbf{0}$)

Calculate $D\mathbf{v}_j = \mathbf{d}_j^v = \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \mathbf{S}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \right)^{-1} \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \mathbf{b}_{ij} \right) (6 \times 1)$ for $j = 1, 2, \dots, N_I$.

Define two $6 \times N_I$ matrices: $\mathbf{D}^v = [\mathbf{d}_1^v \ \mathbf{d}_2^v \ \dots \ \mathbf{d}_{N_I}^v]$ and $\mathbf{V} = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_{N_I}]$

Determine $\mathbf{D}$: $\mathbf{D} = \mathbf{D}^v \mathbf{V}^T (\mathbf{V} \mathbf{V}^T)^{-1} = \frac{6 \cdot d_{11}}{5} \bar{\mathbf{D}}$